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Safety and Treatment Efficacy of Systemic, Locoregional, and Nanoparticle Encapsulated Anthracycline Chemotherapy: Insights Into Cardiotoxicity

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ABSTRACT: Cancer survivors are at an increased risk for major adverse cardiovascular events and cardiovascular disease mortality compared with the general population. In fact, emerging evidence suggests that cardiovascular disease is the leading cause of mortality 10 years into survivorship for some cancer populations. While this heightened cardiovascular risk undoubtedly involves a variety of factors, antineoplastic therapies, such as anthracyclines, are known to contribute significantly to this pathology. As such, modalities that improve the localization of treatments to the tumor while simultaneously minimizing off-target effects on noncancerous cells are greatly needed. Approaches in which anthracyclines are injected directly into tumor-fed arterioles (locoregional/intra-arterial administration) or are encapsulated in tumor-directed nanoparticles have gained prominence in certain cancers. Such therapies should direct cardiotoxic chemotherapy away from sites of potential toxicity while leaving the tumor exposed. However, current clinical findings on the efficacy and cardiovascular safety of these treatments remain inconclusive, and their use for many cancers is limited. This narrative review aims to highlight current clinical findings that have evaluated the safety and efficacy of locoregional and nanoparticle-mediated anthracycline administration as they compare with systemic delivery and identify avenues to be explored by future research.

Key Words: cardio-oncology ■ doxorubicin ■ leukemia ■ neoplasms ■ therapeutics

The antitumor effects of anthracyclines were first described in the early 1960s by Di Marco et al following the isolation of daunorubicin from a strain of *Streptomyces* bacteria.^{1,2} Since then, several anthracycline derivatives, including doxorubicin, epirubicin, and idarubicin, have become staples in the treatment of many childhood and adult cancers. Indeed, lymphomas, leukemias, sarcomas, gynecological cancers, and malignancies of the breast, lung, and liver^{3,4} are all regularly treated with anthracyclines, resulting in an estimated 1 million North Americans treated with these drugs annually.⁵ While this volume underscores their importance as a first-line treatment, the efficacy of anthracyclines, as

well as all other cancer treatments, is heavily dependent on how much of the treatment reaches the tumor. This is determined by a complex interaction of tumor, drug, and treatment-related factors, such as drug dosing and pharmacokinetics, tumor type, size, and feeding vessel characteristics, and the presence of cellular drug resistance pumps (reviewed in the study by Dewhirst and Secomb⁶). Anthracyclines are often delivered systemically via intravenous infusion, and it has been postulated that intratumoral concentrations of doxorubicin are <1% of the administered dose.^{7,8} Indeed, 1 hour following administration of 25 mg of doxorubicin, Stallard et al⁹ report intratumoral concentrations of 220 to 1590 ng/g

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Nonstandard Abbreviations and Acronyms

ABV	adriamycin, bleomycin, and vincristine
BC	breast cancer
CHF	congestive heart failure
CVD	cardiovascular disease
EGFR	epidermal growth factor receptor
HCC	hepatocellular carcinoma
HER2	human epidermal growth factor receptor 2
IA	intra-arterial
ICI	immune checkpoint inhibitor
KS	Kaposi sarcoma
LVEF	left ventricular ejection fraction
ORR	overall response rate
OS	overall survival
TACE	transarterial chemoembolization

of tissue. Likewise, intratumoral doxorubicin concentrations of 3.77 to 19.02 $\mu\text{g/g}$ of tissue have been reported 30 minutes following a bolus injection of 75 mg/m^2 .¹⁰ Studies in rodent models have confirmed that large percentages of viable tumor cells, especially those $>50\ \mu\text{m}$ from a blood vessel, are not exposed to sufficient levels of doxorubicin following single infusions of 8 to 30 mg .¹¹ While increasing the dose of systemic treatments could logically achieve increased tumor delivery, this must be balanced against the potential off-target side effects that can be caused by high systemic drug concentrations. In the case of anthracyclines, these side effects include the potential risk of severe cardiotoxicity as anthracycline use is associated with overt cardiotoxicity (ie, heart failure) in 6% to 10% of treated.^{12,13} Thus, there is a critical need to identify drug delivery techniques that overcome these limitations (Figure 1) and maximize tumor delivery while limiting systemic cardiovascular side effects.

Two promising techniques aimed at improving tumor localization of anthracyclines and reducing systemic side effects include (1) localized delivery to the tumor via direct infusion into tumor-feed arteries (ie, locoregional or intra-arterial [IA] delivery) and (2) encapsulation of treatments into nanoparticles that release the majority of their contents once embedded in tumor tissue (ie, nanomedicine; Figure 2). However, while both locoregional therapy (eg, transarterial chemoembolization [TACE]) and nanoparticle encapsulated anthracyclines (eg, Doxil) have become commonplace in the treatment of some individual cancers (eg, liver, gynecological, and breast), they are not regularly administered for the vast majority of malignancies treated with anthracyclines. This poses the question: if these treatments do achieve these goals, why are they not used universally across various cancers?

Highlights

- Anthracycline chemotherapy is a staple in the treatment of lymphomas, leukemias, sarcomas, gynecological cancers, and malignancies of the breast, lung, and liver.
- Anthracyclines are largely delivered via systemic intravenous infusion, which can result in low intratumoral drug concentrations and off-target side effects, including severe cardiotoxicity.
- These limitations have prompted interest in the use of intra-arterial/locoregional and nanoparticle encapsulated delivery techniques with the goal of improving treatment efficacy and reducing cardiotoxicity.
- Despite growing interest in the use of these techniques, there are limited data on their efficacy and ability to reduce cardiotoxicity compared with systemic anthracycline administration.
- Future work including translational preclinical models and direct comparative studies in cancer populations are critical to elucidate whether intra-arterial/locoregional and nanoparticle anthracycline administration offer substantial improvements in cancer treatment efficacy and cardiovascular safety.

This narrative review seeks to discuss the available evidence on treatment efficacy and cardiovascular safety of locoregional and nanoparticle delivery of anthracyclines compared with systemic administration. Our primary goal was to identify clinical studies that assess the efficacy and cardiotoxicity of locoregional and nanoparticle-mediated delivery of anthracycline chemotherapy and to highlight potential future directions for research within this field. Importantly, this review is not comprehensive, and included studies were identified via a targeted literature search using Google Scholar and PubMed with various combinations of the following keywords: locoregional anthracyclines, IA anthracyclines, anthracycline nanoparticles, TACE, Doxil, systemic anthracyclines, cardiotoxicity, and anthracycline cardiotoxicity. In addition, while anthracycline-based regimens for many cancers are briefly mentioned, the majority of our commentary is centered around breast and liver cancers where comparative studies of systemic and locoregional or nanoparticle-mediated delivery are available. We first discuss the treatment efficacy of systemically administered anthracyclines and briefly overview what is known about anthracycline cardiotoxicity and its mechanisms. Next, we highlight studies comparing the efficacy and incidence of cardiotoxicity between systemic anthracyclines and locoregional and nanoparticle-mediated delivery techniques, before concluding with future directions and needs for additional research within this field. When available, we will direct the interested reader to more detailed reviews on individual topics.

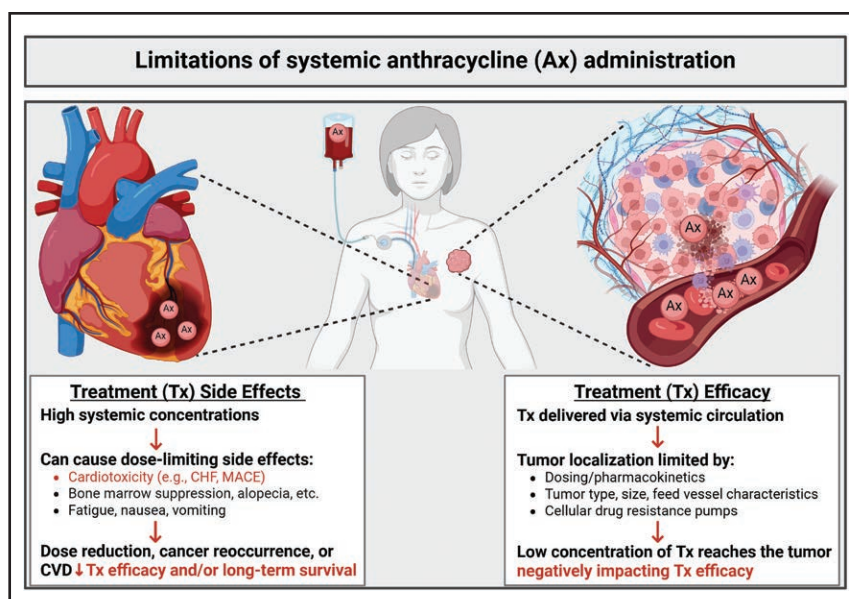


Figure 1. Limitations of systemic anthracycline (Ax) administration.

Systemic administration of Axs is plagued by off-target side effects such as cardiotoxicity and delivers only a small percentage of the administered treatment (Tx; <1%) to the tumor. These limitations not only impede Tx efficacy but also increase the risk of future major adverse cardiovascular events and cardiovascular disease (CVD) during cancer survivorship. In theory, Tx modalities that (1) increase the fraction of the Tx to the tumor and (2) reduce uptake by healthy tissues (eg, heart and arteries) would improve Tx efficacy while reducing the risk of short- and long-term cardiovascular events and mortality. Created with BioRender. Hammond, S. (2026) <https://BioRender.com/sktv33h>. CHF indicates congestive heart failure.

EFFICACY OF SYSTEMICALLY ADMINISTERED ANTHRACYCLINE CHEMOTHERAPY: A FOCUS ON CANCERS NOW TREATED WITH LOCOREGIONAL AND NANOPARTICLE-BASED ANTHRACYCLINES

Soon after its discovery, Di Marco et al¹⁴ demonstrated that doxorubicin exhibited greater antitumor activity than other anthracyclines known at that time (eg, daunorubicin). This finding subsequently led to the use of doxorubicin as a single-agent anticancer therapeutic in clinical settings, and as more clinical data became available, in combination with other antineoplastic drugs. In an early phase II study that included multiple cancer populations, patients showed an overall response rate (ORR) of 25% to doxorubicin.¹⁵ However, the authors found varying success between cancer types (lymphomas 44%, sarcomas 33%, and breast cancer [BC] 32%) with cancer regression evident within 3 treatments. In an extensive review, Blum and Carter¹⁶ compiled the results of the early clinical studies employing doxorubicin as a single agent and reported an ORR of 31% for various cancers. More recent studies assessing doxorubicin as a single agent have reported ORR of 30 to 41 for metastatic BC^{17,18} and 14% to 19% for soft tissue sarcoma.^{19,20} Taken together, doxorubicin as a single agent can lead to tumor regression in ≈15% to 44% patients, depending on cancer type and doxorubicin dosing regimen.

In today's clinical setting, doxorubicin is commonly administered alongside other chemotherapies as part of multidrug regimens that can vary by cancer type, stage, and individual patient factors. These therapies can include (but are not limited to) cyclophosphamide, 5-fluorouracil, paclitaxel, cytarabine, ifosfamide, cisplatin, methotrexate, and vinblastine.^{21–24} For BC, which is the most commonly diagnosed cancer in women, doxorubicin is typically used in conjunction with other chemotherapies. The genetics of cancer play a crucial role in the selection of the drugs used. For example, doxorubicin is used for patients with early stage locally advanced triple-negative BC (negative for progesterone, estrogen, and HER2 [human epidermal growth factor receptor 2] receptors), whereas receptor-specific anticancer treatments (such as trastuzumab for the treatment of HER2-positive BC) are preferentially used in patients with cancers overexpressing any of these receptors. In an early study of patients with advanced BC, combination therapy of doxorubicin and cyclophosphamide resulted in an ORR of 73%.²⁵ In a trial including 1550 patients with BC, the use of adjuvant treatment including doxorubicin, 5-fluorouracil, and cyclophosphamide demonstrated a dose-dependent effect on estimated overall survival (OS) and disease-free survival.²⁶ Patients were assigned to high-, medium-, and low-dose cohorts and received either 60-mg/m² doxorubicin and 600-mg/m² of cyclophosphamide and 5-fluorouracil, or 40 and 400 or 30 and 300 mg/m² of the same drug cocktail. Patients on the high and medium arms had an estimated OS of 79%

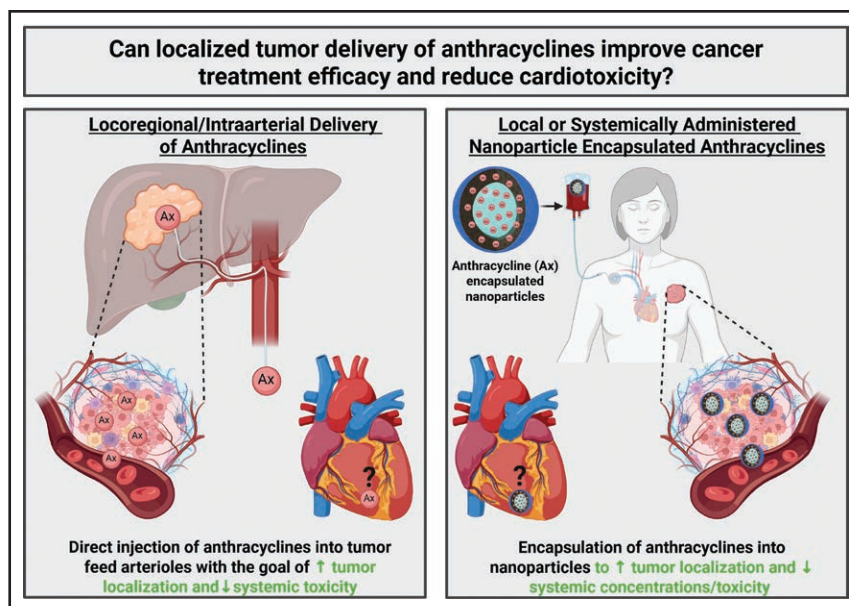


Figure 2. Localized and tumor-directed anthracycline (Ax) delivery as a modality to improve treatment efficacy and reduce cardiotoxicity.

Locoregional or intra-arterial administration of Axs (**left**) has become commonplace in some cancers, such as hepatocellular carcinoma. Direct injection of Axs into tumor-feed arterioles followed by vascular embolization is hypothesized to increase tumor delivery of treatments and minimize systemic circulating Ax levels. However, whether this improves treatment efficacy or reduces cardiotoxicity remains unclear. Encapsulation of Axs in tumor-directed nanoparticles (**right**) has been demonstrated to improve tumor uptake and reduce circulating Ax levels, yet treatment efficacy often remains unchanged compared with conventional treatments. Early findings suggest that this technique may slightly reduce the risk of cardiotoxicity in some cancers (eg, breast cancer); however, more work is needed to determine whether this improves long-term cardiovascular outcomes. If nanoparticle encapsulated Axs result in similar efficacy but reduce cardiotoxicity, this technique may provide an avenue for continued use of Ax treatment with reduced long-term risk to cardiovascular well-being. Created with BioRender. Hammond, S. (2026) <https://BioRender.com/sktv33h>.

and 77% at 5 years, respectively, which was significantly improved compared with patients on the low treatment arm with 72%.²⁶ More recent evidence indicates the addition of taxanes (paclitaxel or docetaxel) to anthracycline and cyclophosphamide-based regimens (with and without 5-fluorouracil) improves disease-free and OS in early stage and operable BC.^{27–29}

In regard to metastatic BC, Miles et al³⁰ reviewed trials evaluating various anthracycline-containing combination therapies, reporting response rates ranging from 40% to 68% when doxorubicin or epirubicin was combined with taxanes or cyclophosphamide. A phase III clinical trial evaluated the efficacy of doxorubicin in combination with vinorelbine compared with doxorubicin as a single agent in patients with metastatic BC.¹⁸ The combination regimen did not result in a significant improvement in ORR (38% versus 30%) and showed no improvement in OS (13.8 versus 14.4 months).¹⁸ With the emergence of targeted therapies, anthracycline use in this setting and in clinical guidelines is becoming less common³¹; however, it is still considered a treatment option,³² particularly in the form of anthracycline-loaded nanoparticles as will be discussed in the Nanomedicine-Mediated Anthracycline Administration section.

In other cancers regularly treated with anthracycline-loaded nanoparticles (ie, AIDS-related sarcoma, multiple

myeloma, and so on; discussed in the Does Nanoparticle-Mediated Administration of Anthracyclines Improve Treatment Efficacy? section), there is limited evidence demonstrating superiority of combining anthracyclines with other therapies compared with treatment with anthracyclines alone. In patients with AIDS-related sarcoma, combining doxorubicin with bleomycin and vincristine (adriamycin, bleomycin, and vincristine [ABV] regimen) significantly improved complete and partial tumor responses compared with doxorubicin alone (ABV: 88% versus doxorubicin: 44%).³³ However, given cancers such as multiple myeloma and recurrent platinum-sensitive ovarian cancer, adding anthracyclines to already established regimens,^{34–36} seemingly little data exists comparing the efficacy of single-agent anthracyclines to combination therapy for these cancers.

ANTHRACYCLINE CARDIOTOXICITY

Introduction to Anthracycline-Induced Cardiotoxicity

Several early reports of anthracycline cardiotoxicity were published between 1967 and 1971, indicating that daunorubicin (also known as daunomycin and rubidomycin) and doxorubicin (also known as adriamycin) may result

in histological abnormalities in cardiac tissue³⁷ or neurons³⁸ and cause congestive heart failure (CHF).^{39–43} Subsequent studies found that this cardiotoxicity was largely dose-dependent, and that 26% to 30% of patients treated with 550 mg/m² of doxorubicin were at risk of experiencing CHF.^{44,45} This finding has largely resulted in a recommended cap of 550 mg/m² on the lifetime dose of doxorubicin.⁴⁶ Despite these efforts, overt cardiotoxicity is still regularly reported in ≈10% of treated patients with subclinical indices afflicting 18% to 59% of patients depending on how the toxicity is defined.^{12,13,47} Additional factors that may predispose patients to heightened risk of cardiotoxicity include preexisting cardiovascular disease (CVD) or risk factors for CVD (eg, hypertension and diabetes), female sex, increased age, socioeconomic status, and concomitant/prior treatment with other cardiotoxic therapies (such as anti-HER2 therapy, eg, trastuzumab) or radiation.^{48–50} Recent data suggest that genetic factors, including single-nucleotide polymorphisms affecting proteins involved in chemoresistance, metabolism, and oxidative stress pathways, may significantly contribute to, or be associated with, the development of doxorubicin cardiotoxicity.^{51–53} For a comprehensive overview of specific single-nucleotide polymorphisms and their associations with doxorubicin-induced cardiac dysfunction, including those conferring reduced risk, the reader is referred to other reviews and original studies where these topics are discussed in detail.^{51,52}

Anthracycline cardiotoxicity has historically been characterized into 1 of 3 categories based on time to disease onset: acute, early onset, or late-onset (reviewed in detail in the study by Giantris et al⁵⁴). Briefly, acute cardiotoxicity is rare (≈1% of cases) and generally occurs within the first week of treatment as a depression of myocardial function, which can rapidly progress to CHF.^{49,54,55} Early onset cardiotoxicity is the most common (≈98% of cases) and comprises chronic dilated cardiomyopathy that arises within 1 year of completing treatment.^{49,54,55} Late-onset cardiotoxicity appears at least 1 year following completion of treatment and is characterized by a progressive reduction in left ventricular function.^{49,54,55} While some originally proposed these may be due to separate disease pathologies, Cardinale et al¹² suggest that these categorizations may simply reflect different stages of the same disease, which are dependent on when/how the disease was identified (eg, via asymptomatic diagnostic imaging in the year following treatment versus waiting until symptom onset) and individual patient factors. Specifically, the authors found that 98% of anthracycline cardiotoxicity cases are detectable as asymptomatic decrements in left ventricular ejection fraction (LVEF) <1 year following completion of treatment (median time to cardiotoxicity=3.5 months). Importantly, 82% of cases were at least partially reversible upon initiation of heart failure treatment.¹² More recent reports have progressed

beyond only the assessment of systolic dysfunction alone, associating anthracyclines with diastolic dysfunction^{56–59} and vascular abnormalities such as endothelial dysfunction^{60–64} and arterial stiffness,^{63,65,66} suggesting that there is still much to learn about the cause and presentation of anthracycline cardiotoxicity. Alterations in vascular function often precede and contribute to the onset of overt CVD, and vascular indices have been shown to be associated with major adverse cardiovascular events or CVD mortality in some cancer populations.^{67,68} Thus, additional work on the potential role of subclinical vascular dysfunction in the progression of anthracycline-induced cardiotoxicity remains an important gap for future study.

Mechanisms of Anthracycline Cardiotoxicity

The underlying mechanisms of anthracycline-induced cardiotoxicity have been frequently studied, yet remain incompletely defined. Herein, we aim to provide only a brief overview of these mechanisms and direct the interested reader to several excellent reviews focused specifically on this topic.^{69–71} In short, the most frequently discussed mechanisms associated with anthracycline cardiotoxicity broadly involve DNA damage, generation of reactive oxygen species, and induction of mitochondrial dysfunction.^{69–72} However, it is worth noting that much of this work has been completed in preclinical rodent or cell culture models with methodologies and dosing regimens that do not always reflect human physiology (ie, tumor naive rodents, intraperitoneal bolus injections versus systemic intravenous injections, and so on).^{71,73}

Anthracyclines primarily mediate their antitumor effects through disruption of topoisomerase II, an enzyme involved in DNA replication and transcription, which prevents DNA repair and causes double strand breaks.^{74,75} Their interaction with this enzyme has also been proposed as a mechanism for cardiotoxicity.^{76,77} Deng et al⁷⁷ demonstrated a protective effect of topoisomerase II β knockdown on the adverse cardiac side effects of doxorubicin. These authors propose dexrazoxane, an iron chelating agent used clinically to treat anthracycline cardiotoxicity,⁵⁰ may prevent doxorubicin cardiotoxicity through modulation of this pathway. Several studies have associated the manifestation of anthracycline cardiotoxicity with the production of reactive oxygen species.⁷⁸ Anthracyclines contain a quinone moiety in their chemical structure that is prone to redox cycling and subsequent generation of superoxide.⁷⁹ Knockout of the superoxide-generating enzyme NADPH oxidase 2 has been sufficient to prevent doxorubicin-induced decrements in myocardial function in some rodent models.⁸⁰ In addition, anthracyclines may also cause reactive oxygen species through interaction with mitochondria. Anthracyclines have an established affinity for cardiolipin,^{81–83} a phospholipid highly expressed in the mitochondrial inner membrane, resulting in

accumulation of these drugs within the mitochondria. This mitochondrial sequestration of anthracyclines can contribute to increased mitochondrial DNA damage,⁸⁴ impaired oxidative phosphorylation,⁸¹ disruption of mitochondrial calcium homeostasis and membrane potential,^{85–87} mitochondrial reactive oxygen species generation,^{88,89} and ultimately cellular senescence or cell death.⁹⁰ Taken together, anthracycline-induced cardiotoxicity is likely driven by a complex interaction of several cellular pathways resulting in oxidative stress and damage to DNA and mitochondria. Interestingly, these are also common traits of aging cells, leading some to propose that anthracycline treatment accelerates cardiac aging.^{71,91,92} This hypothesis is further supported by evidence that patients treated with anthracyclines often present with traits reflected in aged populations, including reduced exercise capacity/muscular strength, impaired vascular endothelial function, and frailty.^{71,91} These findings have prompted some to investigate the effects of various exercise,^{93–97} dietary,^{98,99} or pharmacological interventions^{100,101} aimed at offsetting these side effects and preventing this accelerated aging phenotype in anthracycline-treated patients. To date, many of these trials have had mixed success, and more work is needed to identify adequate countermeasures for preventing anthracycline-induced cardiotoxicity, including both the development of adequate preventative measures and anthracycline delivery techniques, which distribute more drug to the tumor as opposed to healthy cells.

EFFORTS TO IMPROVE TREATMENT EFFICACY AND REDUCE ANTHRACYCLINE CARDIOTOXICITY

Locoregional Chemotherapy Administration

While systemically administered chemotherapy remains an integral component of treatment regimens for many cancers, it is often limited by toxicity to healthy tissue and inability to deliver therapeutic levels of the drug to the tumor.¹⁰² In the 1950s, Klopp et al¹⁰³ sought to remedy this dilemma through local administration of chemotherapy drugs to the region of the tumor via IA infusion. This approach, now coined locoregional or IA therapy, has become commonplace in the treatment of liver cancer^{104,105} and is occasionally used in colorectal and neuroendocrine cancers with liver metastases.^{106–108} Such approaches have also been tested in cancers of the breast,^{109–113} pancreas,^{114–116} prostate,^{117,118} lung,^{119,120} and head/neck^{121,122} but are scarcely used in routine clinical practice, likely due to inconclusive findings on efficacy over systemic treatment and reports of heightened local toxicity.¹¹⁰ In general, these procedures are minimally invasive and involve IA administration of chemotherapeutics or irradiated microspheres into tumor-feeding arterioles. These treatments are often (but not

always) followed by embolization to reduce or block tumor blood flow and maintain high concentrations of the treatment within the tumor while minimizing systemic toxicity.¹²³

Does Locoregional Administration of Doxorubicin Improve Treatment Efficacy?

Locoregional delivery of anthracyclines (primarily doxorubicin) via TACE has become commonplace in the treatment of hepatocellular carcinoma (HCC), as systemically administered doxorubicin has been found to have minimal activity against HCC.¹²⁴ However, a clear consensus on whether this technique, or the use of doxorubicin for HCC in general, improves OS is lacking. While some studies suggest a potential benefit of locoregional doxorubicin in HCC,^{125,126} others have found no survival benefit compared with systemic therapy.¹²⁷ Recent investigations suggest no survival benefit when doxorubicin is used in addition to other common therapies, leading some to question the continued use of doxorubicin in HCC treatment regimens.^{128,129} Few studies have directly compared systemic and locoregional administration of doxorubicin or other anthracyclines. Key available studies are summarized in Table 1. In a small cohort of patients with HCC, Kalayci et al¹²⁷ found that IA administration of doxorubicin+lipiodol did not improve tumor response or median survival compared with intravenous administration (3 versus 4 months for IA versus intravenous, respectively). In a study of 100 patients with HCC comparing locoregional doxorubicin+cisplatin with systemic doxorubicin alone, Mabel et al¹³⁰ found that patients treated with locoregional doxorubicin+cisplatin had significantly higher partial tumor responses (IA: 32% versus intravenous: 16% of patients) and mean progression-free survival (IA: 32 weeks versus intravenous: 26 weeks). However, OS (IA: 38 versus intravenous: 32 weeks) was not statistically different between the groups.¹³⁰ Likewise, in a small cohort of patients with locally advanced BC, IA infusion of epirubicin resulted in significantly higher regression of the primary tumor (IA: 68 versus intravenous: 36%) and reduced side effects compared with those treated with the same concentration of systemic intravenous epirubicin.¹¹² In a trial of 197 patients with inoperable pancreatic cancer, Han et al¹³¹ found improved ORRs (IA: 77.1% versus intravenous: 35.7%) and median survival time (IA: 13.5 months versus intravenous: 6.2 months) following IA administration of doxorubicin, mitomycin, and 5-fluorouracil compared with the same regimen delivered intravenously. While anthracyclines are not commonly used to treat pancreatic cancer, a more recent meta-analysis comparing IA and systemic intravenous administration of various chemotherapy regimens suggests improved partial response rates and fewer complications when IA approaches are used to treat advanced pancreatic cancer.¹¹⁶ In further support of this notion, Mott et al¹³² recently reviewed IA

Table 1. Select Studies Comparing Efficacy of Locoregional/IA Anthracycline Administration With Systemic Anthracycline Administration

Treatment	Population	Design	Outcomes	Reference
Assigned to: TAE/TACE Conventional DOX IV	HCC	Meta-analysis	Primary end point: survival ↑ Survival 2-y survival TAE/TACE: 41% (19%–63%) vs conventional treatment 27% (11%–50%) Secondary end point: ORR ↑ Responsiveness with TACE with cisplatin or DOX vs TAE alone	Llovet et al, 2003 ¹²⁵
Assigned to: TAE (n=37) TACE (gelfoam+DOX; n=40) Conventional DOX IV (n=35)	HCC (n=112)	RCT; prospective measures of arterial embolization DOX, 25–75 mg/m ²	Primary end point: survival Arterial embolization 1 y: 75% 2 y: 50% 3 y: 29% Chemoembolization 1 y: 82% 2 y: 63% 3 y: 29% Conservative treatment. 1 y: 63% 2 y: 27% 3 y: 17%	Llovet et al, 2002 ¹²⁶
Assigned to: TACE DOX+lipiodol (n=18) Conventional DOX IV (n=22)	HCC (n=40)	Prospective measures DOX, 60 mg/m ² every 3–4 wk	Primary end point: survival ↔ Median survival Secondary end point: response rate ↔ Responses rate (IA=11%, IV=14%)	Kalayci et al, 1990 ¹²⁷
Assigned to: TACE (lipiodol, DOX, and cisplatin; n=50) Conventional DOX IV (DOX alone; n=50)	HCC (n=100)	RCT DOX, 60 mg/m ² every 4 wk	Primary end point: survival ↔ Median survival Secondary end point: response rates TACE: ↑ partial response rate 32% vs 16% conventional treatment	Mabed et al, 2009 ¹³⁰
Assigned to: Neoadjuvant therapy (n=26) IA infusion of epirubicin (n=25) IV infusion of epirubicin (n=25)	Advanced BC (n=76)	Prospective randomized study Epirubicin, 20 mg/m ² every 2 wk	Primary end point: survival ↔ Median survival Secondary end point: Response rates IA: ↑ 68.2% vs 36.0% IV	Takatsuka et al, 1994 ¹¹²

↑=improved (survival and response rate) or elevated (incidence of cardiotoxicity). ↔=unchanged (survival, response rate, and incidence of cardiotoxicity). ↓=reduced (survival, response rate, and incidence of cardiotoxicity). BC indicates breast cancer; DOX, doxorubicin; HCC, hepatocellular carcinoma; IA, intra-arterial; IV, intravenous; ORR, overall response rate; RCT, randomized controlled trial; TACE, transarterial chemoembolization; and TAE, transarterial embolization.

approaches for the treatment of head, neck, and central nervous system cancers where they include a table of ≈60 ongoing, recently completed, or ongoing clinical trials investigating IA approaches for the treatment of various malignancies, indicating rising interest in the use of this technique for a variety of cancers. Of the listed trials, 13 included the use of anthracyclines (10 HCC, 2 gastric cancer, and 1 osteosarcoma). To date, results are published for 3 of these trials. In a phase I trial (Unique identifier: NCT02028949), Guiu et al¹³³ completed a dose escalation trial using an idarubicin-lipiodol emulsion without embolization in patients with HCC to assess the efficacy and safety of the procedure. In 15 patients, they found an ORR of 29% 1 month after the second session, a median OS of 20.6 months, and no incidence of cardiotoxicity. The authors report that a phase II trial is scheduled. A phase II study (Unique identifier: NCT02949258) determined the safety and efficacy of IA administration of epirubicin, etoposide, and oxaliplatin followed by oral S-1 (SEEOX regimen) in patients with gastric cancer, reporting that SEEOX is safe and effective in this population, and a phase III trial will be conducted to compare

this regimen to the standard of care.¹³⁴ A phase III trial (Unique identifier: NCT02973685) comparing hepatic arterial infusion of oxaliplatin, 5-fluorouracil, and leucovorin (FOLFOX) to TACE with epirubicin, lobaplatin, lipiodol, and polyvinyl alcohol particles found that the FOLFOX regimen significantly improved OS (FOLFOX: 23.1 months; TACE: 16.1 months) and ORR (FOLFOX: 46% months; TACE: 18%) compared with TACE in patients with unresectable HCC.¹³⁵ Though these findings indicate the potential of nonanthracycline-based regimens to rise in prevalence for HCC treatment, it is important to note that 5-fluorouracil-based regimens are also known to cause cardiotoxicity,^{136–138} thus reaffirming the need for understanding the potential cardiotoxic effects of locoregional and systemic administration of both anthracycline and nonanthracycline-based regimens.

Does Locoregional Administration Reduce Anthracycline Cardiotoxicity?

Aside from the proposed anticancer effects, locoregional administration of anthracyclines is suspected to reduce systemic side effects such as cardiotoxicity.

However, locoregional treatments are most frequently used in liver cancer where cancer recurrence is a primary concern, and cancer mortality remains among the top 5 globally.^{104,139} Considering the low OS rate of patients with these malignancies (5 year relative survival of 22%¹⁴⁰), little is known about the adverse cardiovascular effects of locoregional treatment in this population, especially during long-term follow-up. A recent case report describes severe ventricular dysfunction in an HCC patient treated with locoregional doxorubicin for 1 year.¹⁴¹ Following completion of 6 treatments (5 conventional TACE, 1 drug-eluting nanoparticle TACE, and total cumulative doxorubicin=268 mg/m²), the patient presented with bilateral pleural effusion, increased levels of N-terminal pro-B-type natriuretic peptide, left ventricular dilation, and a severe reduction in LVEF (23%). While the patient received both conventional and nanoparticle-mediated TACE treatments, the authors speculate that the adverse cardiovascular effects were likely a consequence of the conventional TACE, potentially via the release of doxorubicin into the systemic circulation via intrahepatic shunts.¹⁴¹ Farid et al¹⁴² assessed cardiac function via echocardiography 1 day before and 1 month following completion of a single TACE (30-mg/m² doxorubicin hydrochloride) in patients with HCC. Patients were separated into groups based on their response to treatment (ie, complete response, partial response, and stable disease with no response to treatment). Interestingly, there was a modest, albeit statistically significant, decline in LVEF in all groups, while global longitudinal strain was reduced only in patients with complete and partial responses. Likewise, Vogl et al¹⁴³ also demonstrated statistically significant reductions in LVEF following completion of 3 TACE treatments. However, it is important to note that although both studies found statistically significant reductions in these cardiac parameters, the decline in these variables is clinically negligible, the values remain well within the normal ranges for both ejection fraction and global longitudinal strain, and neither variable is reduced to a level suggestive of cardiotoxicity in recent cardio-oncology guidelines.⁵⁰ In addition, these studies are completed acutely with follow-up times ranging between 1 and 4 months, so little is known about the long-term cardiovascular side effects of IA injection of anthracyclines in patients with HCC. Interestingly, a recent SEER database analysis of independent risk factors for CVD death in patients with HCC suggests that those receiving chemotherapy were less likely to die of CVD than those not receiving chemotherapy treatment,¹⁴⁴ thus hinting at potential therapy-independent effects that increase the risk of major adverse cardiovascular events. The authors note that specific chemotherapy information was not available for further analysis, and patients with higher grades who are more likely to receive chemotherapy may have succumbed to HCC before CVD could develop.¹⁴⁴

The efficacy of locoregional anthracycline administration has been assessed in liver metastases from neuroendocrine¹⁰⁷ and colorectal cancers,¹⁰⁶ but, unfortunately, limited data are available on their cardiovascular side effects. Drougas et al¹⁴⁵ report the development of supraventricular tachycardia in 2 patients with metastatic carcinoid tumors following 5 days of IA infusion of various chemotherapy drugs, including 60 mg of adriamycin, but it is difficult to discern a causative effect of adriamycin compared with the other infused drugs (eg, 5-fluorouracil). Thus, as survival rates for liver cancers continue to increase, long-term cardiac follow-up of patients treated with locoregional treatment will be crucial to understanding whether local delivery of anthracyclines to tumors prevents or reduces the adverse cardiovascular side effects associated with systemic infusion.

Despite several studies assessing the efficacy of locoregional infusion of anthracyclines in BC, little data are available on the cardiotoxicity of these regimens, especially as they compare with systemic treatment. Fiorentini et al¹⁰⁹ found no evidence of cardiotoxicity in 36 patients with BC with locally advanced disease following a median of 4 cycles of IA epirubicin and mitoxantrone. Similarly, Pacetti et al¹¹¹ found no evidence of cardiotoxicity in 10 patients with locally advanced BC over the age of 75 years following IA administration of epirubicin, 5-fluorouracil, and mitomycin. However, the authors did note 2 instances of transient cerebrovascular ischemia.¹¹¹ He et al¹¹³ report favorable disease control and tumor response rates in a small sample of patients receiving IA infusion of cyclophosphamide, epirubicin, and 5-fluorouracil compared with systemic delivery of the same drugs. Though no differences in systemic side effects were reported between groups, there was no mention of cardiotoxicity, potentially due to the short duration of follow-up (8–48 months). Thus, there remains a paucity of data about how IA infusion of anthracyclines impacts the cardiovascular outcomes of patients with BC.

Limitations to Locoregional Delivery

To date, locoregional anthracycline administration is primarily limited to liver cancers (primary and neuroendocrine/colorectal metastases) and has yet to be commonly adopted in the treatment of other malignancies. This is largely due to the uniqueness of the hepatic circulation, as hepatocytes receive the majority of their blood supply via the portal vein (75%–83%) as opposed to hepatic arteries.¹⁴⁶ Conversely, liver tumors are primarily perfused via the hepatic artery (90%–100%),¹⁴⁶ thus allowing for localized treatment and embolization of the hepatic artery without compromising blood flow to healthy hepatocytes. Unfortunately, this approach may not be feasible for malignancies in other organs where both tumor and healthy tissue are perfused by the arterial circulation. In such cases, locoregional administration would expose healthy organs to high concentrations of

anthracyclines, whereas embolization would significantly impair tissue blood flow, both of which may contribute to impaired organ function and toxicity. Indeed, in a small sample ($n=4$) of patients with locally advanced BC, local administration of doxorubicin via the internal mammary artery resulted in severe local toxicity in all treated patients.¹¹⁰ Conversely, Kitagawa et al¹⁴⁷ reported no treatment-related complications in a small cohort of patients with stage IIIb BC treated with doxorubicin or epirubicin via the internal mammary or distal subclavian arteries without embolization. While this study and others have reported promising results for local tumor control, locoregional administration may not prevent metastases, presenting another important limitation for use in BC populations.^{147,148} However, as reviewed recently by Mott et al,¹³² there remains interest in the use of IA chemotherapy in the treatment of a variety of neuroendocrine cancers, suggesting interest in the use of this technique outside of liver cancers alone. Additional limitations include marginal improvements in OS, heightened rates of disease recurrence, and failure to maintain adequate drug concentrations in tumor vasculature.^{102,104,149} While this approach was designed to minimize systemic concentrations of administered treatments, some have found that there is no difference in the pharmacokinetic profile between systemic and IA (with and without lipiodol) administration of doxorubicin.¹⁵⁰ Furthermore, embolization procedures designed to isolate treatments within the tumor may contribute to a hypoxic tumor environment that promotes angiogenesis and cancer cell proliferation.¹⁴⁹

Many of the studies assessing the efficacy of IA treatments provide limited information on the cardiovascular side effects following treatments. Future work comparing systemic and locoregional administration of anthracyclines might consider including cardiovascular measurements. If the efficacy of IA and systemic treatments is similar, but the incidence of cardiotoxicity (and other side effects) is reduced following IA infusion, such treatments may allow for continued use of anthracyclines with minimal burden to cardiovascular health. Nonetheless, the above limitations of locoregional/IA treatments have prompted the study of nanomedicine, where cancer therapies are encapsulated in tumor-targeted organic or inorganic nanoparticles,¹⁵¹ as a potential modality to improve tumor responses, reduce systemic drug concentrations, and minimize cardiotoxicity.

Nanomedicine-Mediated Anthracycline Administration

Like locoregional administration of chemotherapy, nanoparticle-based medicine was developed as a promising approach to improve tumor localization of treatment and reduce systemic side effects associated with chemotherapeutics. Nanomedicine can be defined as the use of molecularly heterogeneous polymers in nanoscale

to deliver medications to a site of interest.^{152,153} There are 2 underlying goals of these delivery systems: (1) to increase treatment efficacy and (2) to avoid general side effects by tumor-directed drug delivery and, thus, improve a patient's OS.¹⁵³ Despite these promising conjectures, only a few nano-based systems have been approved by the Food and Drug Administration.¹⁵² To date, nanomedicine has been demonstrated to fulfill one of its promising goals by increasing the concentration of chemotherapeutics in human cancer biopsies^{154,155} and malignant effusions.¹⁵⁶ Furthermore, it has been demonstrated that these drug-loaded nanoparticles preferentially accumulate in tumors rather than in healthy tissue, supporting the safety of nanoparticles.^{157,158} The current state of nanoparticle-based drug delivery in cancer is largely based on the abnormal characteristics of tumor blood vessels, where the leaky vasculature of tumors allows nanoparticles to extravasate through the vessel wall and accumulate in tumors, a principle called the enhanced permeability and retention effect.^{159,160} However, whether this improved tumor delivery of nanoparticle-based delivery (with enhanced permeability and retention effect) results in heightened treatment efficacy and OS remains unestablished.

Does Nanoparticle-Mediated Administration of Anthracyclines Improve Treatment Efficacy?

A variety of clinical trials have investigated the efficacy, safety, and OS of different nanoparticle-based therapies. Due to the highly varying composition and modifications of nanoparticles used in these studies, which encompass simple liposomal designs to modified vehicle constructs and different active substances (daunorubicin or doxorubicin), various terminologies (eg, pegylated lysosomal doxorubicin and drug-eluting bead-TACE) are used throughout the literature. Herein, we will collectively refer to all nanoparticle-mediated anthracycline compounds as nanoparticles for simplicity and seek to discuss their efficacy and safety compared with systemic and locoregional administration. The efficacy and cardiovascular safety findings of key studies are summarized in Table 2.

In patients with metastatic BC, 3 different randomized phase III trials tested the efficacy of nanoparticle doxorubicin compared with systemic doxorubicin and found comparable ORRs. In 2 studies, systemic doxorubicin and nanoparticle doxorubicin resulted in identical ORRs of 26%¹⁶¹ and 43%.¹⁶² In a third study, Harris et al¹⁶¹ found response rates of 33% and 38% for nanoparticle doxorubicin and systemic doxorubicin, respectively.¹⁶³ Importantly, these response rates were not statistically different in any of the 3 studies. In addition, the OS was not different in any of the 3 studies (nanoparticle versus doxorubicin: 16 versus 20 months,¹⁶¹ 19 versus 16 months,¹⁶² and 21 versus 22 months¹⁶³). Similar findings have been demonstrated in patients with stage I-III BC. Hu et al¹⁶⁴ found no difference in OS or disease-free

Table 2. Select Studies Assessing Efficacy and Cardiotoxicity of NP-Mediated Anthracycline Delivery Compared With Systemic and Locoregional Administration

Treatment	Population	Design	Outcomes (efficacy and cardiotoxicity)	Reference
Systemic anthracycline vs NP anthracycline-based therapy				
Assigned to: Liposome-encapsulated DOX (TLC D-99; n=108) Conventional DOX (n=116)	Meta-static BC	Randomized phase III trials DOX, 75 mg/m ² every 3 wk	Primary efficacy end point: Response rate ↔ 26% in either group Primary safety end point: Cardiotoxicity ↓ NP: 13% vs 29% Survival: ↔ 16 vs 20 mo	Harris et al, 2002 ¹⁶¹
Assigned to: Liposome-encapsulated DOX (Myocet)+ cyclophosphamide (n=142) Conventional DOX+ cyclophosphamide (n=155)	Meta-static BC	Randomized multicenter trial DOX, 60 mg/m ² every 3 wk	Primary efficacy end point: Response rate ↔ 43% in either group Primary safety end point: Cardiotoxicity ↓ 6% vs 21% Median cumulative dose at cardiotoxicity onset: NP: ↑ 2220 mg/m ² vs 480 mg/m ² Survival: ↔ 19 vs 16 mo	Batist et al, 2001 ¹⁶²
Assigned to: Pegylated liposomal DOX (n=254) Conventional DOX (n=255)	Meta-static BC	Randomized phase III trials NP-DOX, 50 mg/m ² every 4 wk DOX, 60 mg/m ² every 3 wk	Primary efficacy end point: Response rate ↔ NP: 33% vs 38% Primary safety end point: Cardiotoxicity, LVEF-based NP: ↓ 3.9% vs 18.8% Survival: ↔ 21 vs 22 mo	O'Brien et al, 2004 ¹⁶³
Assigned to: Pegylated liposomal epirubicin (n=661) Conventional epirubicin (n=810)	Stage I–III BC	Retrospective study Epirubicin, 35–40 mg/m ² every 3 wk	Survival ↔ survival NP 3 y: 98% 5 y: 94.7% 10 y: 94.7% Epirubicin 3 y: 98.2% 5 y: 96.8% 10 y: 89.5%	Hu et al, 2021 ¹⁶⁴
Assigned to: Pegylated liposomal DOX+cyclophosphamide (n=76) Epirubicin+cyclophosphamide (n=78)	Metastatic BC	Phase III trials NP-DOX 75 mg/m ² every 3 wk Epirubicin 75 mg/m ² every 3 wk	Survival ↔ survival NP 18.3 vs 16.0 mo Response rate ↔ NP 46% vs 39% Cardiotoxicity, LVEF-based ↔ NP 11.8% vs 10.2%	Chan et al, 2004 ¹⁶⁵
Assigned to: Pegylated liposomal DOX+vincristine and dexamethasone (n=29) Conventional IV DOX+vincristine and dexamethasone (n=31)	Acute lymphoblastic leukemia (n=60)	Phase II, randomized trial DOX IV, 12 mg/m ² every day for 4 d, repeated 1 mo after NP-DOX, 40 mg/m ² once, repeated 1 mo after	Response rate (complete remission) ↔ NP 72% vs 90% Overall survival ↔ NP 9 vs 10 mo Overall survival at 2 y ↔ NP 24% vs 35% Cardiac events ↔ NP 1 in 29 patients (3%) vs 6 in 31 patients (21%)	Hunault-Berger et al, 2011 ¹⁶⁶
Locoregional anthracycline vs NP anthracycline-based therapy				
Assigned to: DEB-TACE (n=45) TACE (n=21)	HCC (n=71)	Retrospective study DEB-TACE: 75-mg DOX TACE: 50-mg DOX, 100-mg cisplatin, 10-mg mitomycin, and 1:1 emulsion with lipiodol	Primary outcome: median survival DEB-TACE: 610 d TACE: 284 d	Dhanasekaran et al, 2010 ¹⁶⁷
Assigned to: DEB-TACE TACE	HCC (n=700)	Meta-analysis (2014)	Response rate: ↑ DEB-TACE vs TACE Survival: ↑ DEB-TACE vs TACE at 1 and 2 y	Huang et al, 2014 ¹⁶⁸
Assigned to: DEB-TACE (n=822) TACE (n=1010)	HCC (n=1832)	Meta-analysis (2017)	Response rate: ↔ DEB-TACE vs TACE Survival: ↑ DEB-TACE vs TACE at 1, 2, and 3 y	Chen, et al, 2017 ¹⁶⁹

↑=improved (survival and response rate) or elevated (incidence of cardiotoxicity). ↔=unchanged (survival, response rate, and incidence of cardiotoxicity). ↓=reduced (survival, response rate, and incidence of cardiotoxicity). BC indicates breast cancer; DEB, drug-eluting bead; DOX, doxorubicin; HCC, hepatocellular carcinoma; LVEF, left ventricular ejection fraction; NP, nanoparticle; RCT, randomized controlled trial; TACE, transarterial chemoembolization; and TAE, transarterial embolization.

survival at 46-month follow-up in patients receiving adjuvant therapy of nanoparticle doxorubicin with cyclophosphamide and docetaxel/paclitaxel compared with systemic epirubicin coupled with the same complementary chemotherapeutics.

The aforementioned high levels of recurrence and mortality in liver cancer (see the Does Locoregional Administration Reduce Anthracycline Cardiotoxicity? section) have prompted studies to determine whether nanoparticles are effective at treating HCC or can improve upon the efficacy of locoregional doxorubicin administration via TACE. In 2 HCC studies evaluating the efficacy of systemic nanoparticle doxorubicin, little to no response rate with doses of 40 and 30 mg/m² was observed.^{170,171} Both studies had low patient numbers and relatively short OS time with 130 and 140 days, respectively. Varela et al¹⁷² assessed the pharmacokinetic profiles of conventional and nanoparticle TACE and found that nanoparticle TACE significantly reduced serum doxorubicin c_{max} (78.97±38.3 versus 895.66±653.1 ng/mL) and area under the curve (662.6±417.6 versus 1532.98±295.2 ng/mL·min) levels compared with conventional TACE, suggesting reduced passage of doxorubicin into the systemic circulation. However, whether this ultimately correlates with improved efficacy of doxorubicin treatment remains inconclusive. Dhanasekaran et al¹⁶⁷ found that patients with HCC treated with nanoparticle TACE had significantly longer median survival times compared with those treated with conventional TACE (610 versus 284 days). This finding was partially supported by a meta-analysis conducted by Huang et al,¹⁶⁸ which demonstrated significantly higher objective tumor response and 1- and 2-year survival rates in nanoparticle TACE compared with conventional TACE. However, 6-month and 3-year survival rates were not different between groups.¹⁶⁸ Lammer et al¹⁷³ did not detect a statistical improvement in complete (TACE: 22% versus nanoparticle TACE: 27%) or objective tumor responses (TACE: 44% versus nanoparticle TACE: 52%) or disease control (TACE: 52% versus nanoparticle TACE: 63%) between conventional and nanoparticle TACE-treated patients at 6 months despite a trend toward beneficial response following nanoparticle TACE. In a more recent multicenter randomized controlled trial, Golfieri et al¹⁷⁴ found no difference in local or overall tumor response in the first 12 months post-treatment or the 1-year (TACE: 83.5% versus nanoparticle TACE: 86.2%) and 2-year (TACE: 55.4% versus nanoparticle TACE: 56.8%) survival rates between conventional and nanoparticle TACE groups. The opposing findings between these studies may be due to the severity of the disease in the population studied, differing drug combinations (eg, epirubicin versus doxorubicin and with/without cisplatin), and the size and composition of the nanoparticle used. Given the conflicting data on the benefit of anthracyclines in patients with HCC, future approaches may favor the use

of emerging therapies such as sorafenib¹²⁸ or yttrium-90 radioactive beads¹⁷⁵ or use a combination of systemic and targeted nanoparticle treatments to improve HCC survival outcomes.¹⁰⁸

Nanoparticle doxorubicin has also been tested for use in AIDS-related Kaposi sarcoma (KS). One clinical trial testing nanoparticle doxorubicin versus systemic therapy reported an improved OS with 45% for the nanoparticle-treated arm versus 25% in systemically treated patients receiving doxorubicin, bleomycin, and vincristine (ABV and doxorubicin referred to as adriamycin).¹⁷⁶ On the contrary, no difference in ORR was reported by others (25% in nanoparticle and 28% for ABV-receiving patients).¹⁷⁷ Of note, the former study utilized doxorubicin as the active compound, whereas the latter used daunorubicin, which may explain the discrepancy in the ORR. In addition to the different active drugs used, the former study utilized equal concentrations of doxorubicin with 20 mg/m² per 2 weeks for both nanoparticle and ABV arm with the vast majority of patients being men,¹⁷⁶ whereas the latter trial had a more heterogeneous population with varying concentrations of the drug tested.¹⁷⁷ Despite these different parameters used to treat AIDS-related KS, the median OS between nanoparticle and systemically treated patients did not differ in either treatment at 5 months or 1 year, respectively.^{176,177} In another study in which patients with KS had already received systemic doxorubicin-containing ABV chemotherapy and either failed to respond to the treatment, had noted disease progression, or had treatment discontinued due to toxicities, patients were subsequently treated with nanoparticle doxorubicin. Out of the 28 patients who were unresponsive to their prior treatment, 32% had a partial response after treatment with 20-mg/m² nanoparticle doxorubicin.¹⁷⁸ Thus, while nanoparticle doxorubicin may improve OS and tumor response in some cancers, the majority of studies to date suggest no improvement in treatment efficacy over conventional treatments.

Does Nanoparticle-Mediated Administration of Anthracyclines Reduce Anthracycline Cardiotoxicity?

While nanoparticle doxorubicin is used in the treatment of ovarian cancer, multiple myeloma, and AIDS-related KS, the vast majority of available data on its cardiovascular safety have been conducted in BC populations. The variables used to assess cardiotoxicity vary by study, but most determine how treatments impact LVEF through the progression to overt heart failure. In the most comprehensive meta-analysis to date, Cavalli et al¹⁷⁹ analyzed findings from 12 randomized controlled trials of multiple cancers (6 BC, 2 multiple myeloma, 2 lymphoma, 1 sarcoma, and 1 leukemia), which investigated the incidence of cardiotoxicity between nanoparticle and other anthracycline-based regimens. Their findings suggest that nanoparticle-based regimens significantly reduce

heart failure incidence and reduce the clinical decline in LVEF by 61% compared with standard anthracycline-based regimens but do not impact OS or tumor response.¹⁷⁹ Interestingly, when analyses were completed for individual cancer types, reductions in cardiotoxicity in nanoparticle-treated groups were maintained, while no benefit was found for lymphoma and multiple myeloma.¹⁷⁹ In most BC studies, nanoparticle-containing regimens have shown clear superiority over systemic treatments in regard to cardiotoxicity. In patients with metastatic BC, Harris et al¹⁶¹ documented that only 13% of patients with cancer treated with nanoparticle doxorubicin developed cardiotoxicity versus 29% of patients receiving systemic doxorubicin. This trend was confirmed by Batist et al¹⁶² who found cardiotoxicity in only 6% of patients treated with nanoparticles compared with 21% in the conventional systemic therapy group. Similar findings were published by O'Brien et al¹⁶³ with cardiotoxicity incidence rates of 4% and 19% for nanoparticle and conventional doxorubicin treatment, respectively. Likewise, all 3 studies came to the agreement that fewer patients on nanoparticle treatment develop CHF. Only 2 nanoparticle-treated patients developed CHF in the study by Harris et al,¹⁶¹ whereas 9 cases of CHF were observed in the systemic treatment arm. Similarly, both Batist et al¹⁶² and O'Brien et al¹⁶³ found no incidences of CHF in nanoparticle doxorubicin-treated patients but reported 5 and 2 cases in the systemic treatment arms, respectively. Notably, these reduced incidences of cardiotoxicity occurred despite a higher mean cumulative dose of doxorubicin in patients treated with nanoparticle compared with conventional systemic therapy (nanoparticle doxorubicin: 785 mg/m² versus conventional doxorubicin: 570 mg/m²).¹⁶¹ Moreover, nanoparticle-treated patients had a higher median lifetime cumulative dose of doxorubicin at the onset of the first cardiac event (2200 versus 480 mg/m²).¹⁶² In patients with stage I-III BC treated with nanoparticle doxorubicin, Hu et al¹⁶⁴ demonstrated a lesser drop in LVEF compared with those treated with conventional systemic treatment. However, the authors only report the relative change in LVEF for nanoparticle doxorubicin (5.5%) versus conventional epirubicin (6.7%) without indicating what the baseline/final values for LVEF were. Given that these reductions in LVEF are <10% denoted as cardiotoxicity in the European Society of Cardiology cardio-oncology guidelines,⁵⁰ coupled with the lack of information on absolute baseline and posttreatment LVEF values, it is unclear whether these alterations in LVEF indicate clinically significant pathology. In addition, the magnitude of difference between the groups (1.2%) is relatively small and may not represent a clinically significant difference in LVEF. Thus, more work is needed to ascertain whether the use of nanoparticle doxorubicin improves the cardiotoxicity profile compared with conventional epirubicin treatment. In patients with a high risk of BC (elevated risk of cancer

recurrence), no differences were observed with respect to abnormal LVEF and frequency of CHF in systemically and nanoparticle-treated patients.¹⁸⁰ Only systemically treated patients had significantly higher frequency of elevated troponin levels compared with nanoparticle arm, a known marker for cardiotoxicity.^{181,182}

In a study on AIDS-related KS comparing nanoparticle doxorubicin to ABV, the authors did not observe any cardiotoxicity, likely due to the low doses of doxorubicin administered.¹⁷⁶ A similar outcome was observed in a study administering daunorubicin-containing nanoparticles, in which no patients, irrespective of treatment arm, had a drop of LVEF >20%.¹⁷⁷ The authors reported minor cardiac events (eg, palpitations, hypertension, and tachycardia) in 7% of nanoparticle-treated patients and 9.9% of ABV-treated patients.¹⁷⁷

Limitations to Nanoparticle Delivery Systems

Although nanoparticle-based delivery systems in the discussed phase III metastatic BC trials have fewer signs of drug-induced cardiotoxicity, they strikingly failed to achieve the primary goal of oncology nanomedicine, prolonging the OS of patients with cancer.¹⁵³ Similarly, the OS in other cancers is mixed (HCC) or remains unchanged (BC and AIDS-related KS). These surprising observations may serve as an explanation why nanomedicine is not the preferred delivery system clinically, despite showing promise in respect to attenuating anthracycline cardiotoxicity and reducing systemic drug concentrations. These findings are intriguing as they suggest that if patients on anthracycline-based nanoparticles are not severely affected by anthracycline cardiotoxicity but have the same mortality rate as systematically treated patients with cancer, there are other factors in place, which account for the unchanged OS, most likely the primary cancer. In addition, the findings that nanoparticle-treated patients exhibit a lower incidence of cardiotoxicity warrant further discussion. The vast majority of anthracycline cardiotoxicity is detectable within 1 year following the completion of the anticancer treatment,¹² as discussed above (see the Introduction to Anthracycline-Induced Cardiotoxicity section). With the survival of patients with metastatic BC treated with nanoparticles ranging from 16 to 21 months,¹⁶¹⁻¹⁶³ this indeed falls within the range where fewer cases of anthracycline cardiotoxicity were observed. However, these patients with cancer do not survive long enough to develop long-term major adverse events, which still accounts for the high cardiovascular mortality in cancer survivors (>2 years).⁴⁸ While the lack of early cardiotoxicity does give a good indication that fewer long-term events may occur, studies with long-term follow-up are needed to determine whether nanomedicine is sufficient to reduce or even avert cardiotoxicity-induced mortality. Given that the majority of nanomedicine-based studies have been conducted in

metastatic BC and liver cancer populations with shorter OS periods, future work is needed in populations with longer OS to truly determine whether nanoparticle-based therapies can reduce cardiovascular mortality incidence rates in cancer populations.

The failure of these nanoparticle delivery systems to improve OS of patients with cancer may be attributed to multiple underlying mechanisms, which include the heterogeneity of tumor vasculature, leading to uneven drug distribution and, thus, limited intratumoral drug concentration, the improper intracellular trafficking of lysosomes containing liposomes, leading to the sequestration of drug from the nucleus, and the insufficient release of therapeutic drug from lysosomes.¹⁸³ To address these limitations, strategies that incorporate targeting moieties (eg, ligands, antibodies, and peptides) on nanoparticle surfaces have been developed in attempt to bind proteins overexpressed on tumor cells and induce receptor-mediated endocytosis and enhanced uptake of nanoparticle.¹⁵⁹ One such approach was MM-302, an HER2-targeted antibody-liposomal doxorubicin conjugate, which showed a promising safety profile and efficacy when administered either as monotherapy or in combination with trastuzumab.¹⁸⁴ Based on the favorable outcomes obtained from that study, the authors proposed further evaluation of MM-302 in phase II clinical studies; however, these studies were terminated by the sponsor because these targeted nanoparticle systems did not improve treatment efficacy (Unique identifier: NCT02213744).¹⁸³ Another example is C225-immunoliposomal-doxorubicin (Unique identifier: NCT01702129), an EGFR (epidermal growth factor receptor)-targeted antibody-liposomal doxorubicin conjugate tested in patients with various solid tumors.¹⁸⁵ Early trials demonstrated that C225-immunoliposomal-doxorubicin was well tolerated, without cardiotoxicity¹⁸⁵ while pharmacokinetic analyses found significantly reduced plasma half-life of doxorubicin compared with free doxorubicin.¹⁸⁶ However, in a phase II trial (Unique identifier: NCT02833766), C225-immunoliposomal-doxorubicin failed to improve treatment efficacy in patients with triple-negative BC, resulting in early termination of the trial and the conclusion that the treatment should not be further developed for triple-negative BC.¹⁸⁶ Collectively, targeted nanoparticle systems have been developed but have not progressed beyond phase I and II trials and have not achieved broad clinical approval.¹⁵⁹ Therefore, passive nanoparticle, such as Doxil and Myocet, remain the best systemic nanoparticle delivery systems available, and more work is needed to improve the treatment efficacy of these delivery systems.

An additional consideration for the further development of anthracycline-mediated treatments is the emergence of immune checkpoint inhibitors (ICIs). Since the first Food and Drug Administration approvals in 2011 for melanoma and lung cancer, the number of ICIs and

the spectrum of malignancies they treat have expanded substantially; this includes use for cancers traditionally treated with anthracyclines.¹⁸⁷ Thus, while the need for systemic/nanoparticle-based therapies and their development may, at first, seem less urgently required, ICIs are not yet approved or deemed effective against all cancer types, and ICI resistance remains a well-documented challenge that can render these agents ineffective in a subset of patients.^{187,188} Therefore, continued research on not only ICIs but also alternative approaches, including locoregional and nanoparticle-mediated delivery of anthracyclines, remains imperative for improving efficacy and cardiovascular safety of cancer treatments.

FUTURE DIRECTIONS AND CONCLUSIONS

Cancer treatment involves a complex balance of maximizing therapeutic efficacy while minimizing the effects of drugs on healthy tissue. While advances in anticancer treatments have undoubtedly improved the former point, the emerging evidence that cancer survivors are at an increased risk for major adverse cardiovascular events and mortality^{48,189,190} has prompted exploration for how to improve the latter. Specifically, many chemotherapies, especially anthracyclines, are associated with dose-limited cardiotoxicity that can restrict their use. In an attempt to both optimize tumor localization and reduce systemic toxicity, modalities such as IA/locoregional injection and encapsulation of treatments in tumor-directed nanoparticles have risen in interest. Herein, we sought to overview the existing literature on whether these approaches improve treatment efficacy and reduce the incidence of cardiotoxicity. Locoregional therapies have become commonplace in cancers such as HCC; however, their current impact on cancer survival outcomes remains unclear, and data on cardiovascular outcomes are limited. Similarly, while encapsulation of anticancer drugs in tumor-directed nanoparticles can reduce systemic circulating levels of anthracyclines, presumably by isolating treatments within the tumor, OS remains similar to standard treatment options. This inconclusive evidence for improving treatment efficacy has limited the use of locoregional and nanomedicine-mediated approaches to a small subset of malignancies, and thus, there remain insufficient data to draw final conclusions on whether these techniques should be implemented universally.

Nonetheless, the potential of these techniques to sequester treatments within the tumor and reduce cardiotoxicity in some populations is promising, and thus, both approaches remain a focus of several ongoing clinical trials. Note that limitations of these techniques include failure to isolate drugs within tumor tissue, heightened local toxicities, cost of administration, limited longitudinal data outside of 3 years post-administration, and Food and Drug Administration approval. As such,

continued research in this space is warranted, and there remain several exciting avenues (both preclinical and in patients) for research on the use of these approaches to improve cancer care and methods to advance understanding of and reduce anthracycline cardiotoxicity in general (Figure 3).

Future Preclinical Approaches for the Study of Anthracycline Cardiotoxicity

Historically, most preclinical studies in cardio-oncology have utilized nontumor-bearing rodents and anthracycline dosing routes (ie, intraperitoneal) that are dissimilar to clinical practice.⁷³ While these early studies were important for highlighting the independent effects of cancer treatments on the cardiovascular system, future work, including clinically relevant dosing regimens and

tumor-bearing models, is needed to improve the translatability of preclinical findings to the clinical setting and to clarify how the tumor itself contributes to cardiotoxicity. Many prior studies using tumor-bearing rodents, particularly as it relates to the cardiotoxicity of anthracycline-loaded nanoparticles versus free anthracyclines, have used ectopic tumors rather than orthotopic tumors.^{191,192} This is an important point, given that blood flow to ectopic tumors is severely reduced (at rest and during exercise) compared with orthotopically implanted tumors,¹⁹³ and thus, such studies may not accurately reflect the distribution of nanoparticles in human cancer. Those that have used orthotopic models have limited functional cardiac data to assess cardiotoxicity,¹⁹⁴ leaving a gap in knowledge about the mechanistic and functional development of cardiotoxicity in these models. Given the large differences between humans and rodents in physiology

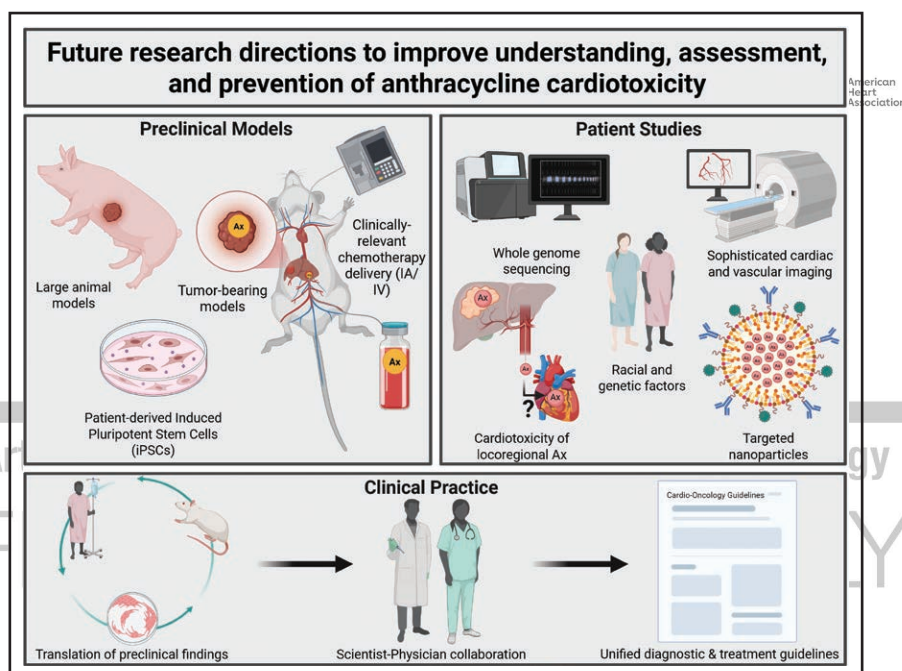


Figure 3. Future directions to improve understanding, assessment, and prevention of anthracycline (Ax) cardiotoxicity.

Locoregional and nanoparticle (NP)-mediated delivery of Axs has aimed to improve treatment efficacy and reduce the cardiotoxicity of Ax-based treatments. Yet, these approaches have yielded mixed findings, and neither has been adopted universally in the treatment of cancer. Future preclinical studies (**top left**) may seek to improve the mechanistic understanding of cardiotoxicity through the use of large animal models, which more closely recapitulate human physiology compared with rodents. In addition, more cardio-oncology studies completed in orthotopic tumor-bearing models and using chemotherapy delivery methods, which closely reflect those seen clinically (including locoregional and NP administration), would provide a better understanding of how the tumor impacts Ax cardiotoxicity and improve translatability of findings to the clinical setting. Analysis of patient-derived induced pluripotent stem cells (iPSCs) and whole genome sequencing of patient samples may assist in the identification of cellular pathways and genetic factors involved in the development of cardiotoxicity in different populations and ethnic groups. Advances in sophisticated imaging (cardiac magnetic resonance imaging and PET-CT) allow for sensitive detection of cardiac and vascular function, and the use of these techniques in human patients (**top right**) undergoing treatment with Axs could help with the detection of Ax-induced cardiovascular maladaptation before irreversible cardiotoxicity develops. There remain limited data on the cardiovascular safety of locoregional treatments, and future work is needed to determine how this delivery technique impacts cardiovascular function. In addition, several clinical trials are currently assessing the clinical utility of nanoparticles modified with antibodies for various receptors that are commonly expressed in cancer cells (eg, HER2 [human epidermal growth factor receptor 2]), and it remains unknown whether this approach will improve efficacy and further reduce cardiotoxicity compared with other Ax treatment modalities. Ultimately, scientists and physicians must continue to work together to translate findings from both preclinical and patient studies to the clinical setting (**bottom**) to facilitate the continued advancement of diagnostic, treatment, and preventive criteria to ensure that patients can continue to receive Axs with minimal risk to short- and long-term cardiovascular health. Created with BioRender. Hammond, S. (2026) <https://BioRender.com/sktv33h>. IA indicates intra-arterial; and IV, intravenous.

in general, future approaches should also consider the use of large animal models, such as swine, which are more reflective of human cardiovascular physiology.¹⁹⁵ In addition, nonanimal models, including human induced pluripotent stem cells, are becoming more commonplace and have the potential to provide a unique preclinical perspective into genetic and molecular pathways involved in anthracycline cardiotoxicity.^{196–198}

Future Patient and Clinical Approaches for the Study of Anthracycline Cardiotoxicity

As highlighted above, there is a paucity of studies investigating whether locoregional administration of anthracyclines reduces cardiotoxicity compared with systemic administration. While locoregional approaches are primarily used in the treatment of liver cancers where high cancer mortality rates may limit potential cardiotoxic effects from being realized, such studies will have implications for understanding long-term cardiovascular health as cancer survival rates continue to improve. Ultimately, a direct comparison of all treatment strategies is necessary to finalize judgment on which approach is the safest and most efficacious for any given cancer. A treatment option that is safe in patients with BC may not be the best choice for HCC or other malignancies. Executing clinical trials with direct side-by-side comparison represents the gold standard, but this approach is costly and time-consuming, and can contain built-in biases, such as a lack of diverse study populations and failure to account for geospatial impact or social determinants of health. Studies integrating whole genome sequencing have the potential to identify heightened risk for cardiotoxicity and improve understanding of the mechanisms for how cardiotoxicity develops across different populations, ethnic groups, and genetic ancestry.^{199,200} Along these lines, there remain limited prospective studies investigating the incidence of anthracycline cardiotoxicity in different racial and ethnic populations, despite retrospective analyses, suggesting that differences may exist.^{201–203} This scarcity of evidence may be, in part, due to the lack of diversity in clinical trials.²⁰² Prospective studies including diverse study populations are important for future consideration of treatment options and implementation of preventative measures. Similarly, potential sex differences in the incidence of cardiotoxicity warrant future study. Preclinical models often cite that male rats have an increased risk for anthracycline cardiotoxicity (reviewed in the study by Meiners et al²⁰⁴), whereas human trials have been less clear with some indicating higher rates in women,²⁰⁵ men,²⁰⁶ or no difference between sexes²⁰⁷ (reviewed in the study by Wilcox et al²⁰⁸). This discordance likely involves a variety of factors, including the age of first exposure to anthracyclines, cancer type, and whether female patients are premenopausal or postmenopausal at the time of study.²⁰⁸ With the rise of national health registries

and utilization of electronic health records globally, we now have the means to answer some of these questions (ie, sex and racial differences) without performing direct comparative clinical trials. Future research utilizing electronic health records in combination with whole genome sequencing (eg, All of Us, UK Biobank, and FinnGen) allows researchers to understand which treatment strategy has the best chance of treating the cancer itself (without increasing risk of adverse cardiovascular events) while simultaneously learning how different populations/genetic predispositions play a role in clinical pathology. Importantly, performing this line of investigation on a global level should eliminate any country or societal-specific predispositions that may be an artifact of access to health care (eg, due to different degrees of regulatory approval in different countries) or social determinants of health that differ due to single-payer health systems versus mostly privatized insurance models. In addition, further development and investigation into emerging treatments, such as modified nanoparticles targeted to proteins highly expressed in various cancers, will also be important to increase the delivery of anthracyclines to tumors as opposed to healthy tissue. When assessing the cardiovascular safety of emerging (and existing) treatments, investigators may consider including additional measures of cardiac and vascular function rather than measurement of LVEF alone. To date, alterations in LVEF have been most frequently used to assess cancer treatment-induced cardiotoxicity. However, subclinical cardiotoxicity may arise before reductions in LVEF and can be detected using measurements of diastolic function and strain.^{209,210} Measurement of global longitudinal strain before and throughout cardiotoxic chemotherapy is now recommended in the European Society of Cardiology cardio-oncology guidelines due to its ability to predict left ventricular dysfunction.⁵⁰ Furthermore, use of risk scoring matrices for heart failure with preserved ejection fraction, which is likely underdiagnosed in cancer populations,²¹¹ would offer an easy-to-implement early warning sign for cardiotoxicity.^{212,213} Given the importance of early detection in preventing irreversible cardiotoxicity,¹² utilization of such measurements and techniques before and following administration of systemic, locoregional, and nanoparticle-based therapies may provide heightened insight into the cardiovascular impact of these treatment modalities and allow for early intervention before irreversible cardiotoxicity develops. In addition, mounting in vivo and ex vivo evidence suggests that numerous cancer treatments, including anthracyclines, may impair macrovascular and microvascular functions.^{60,61,137,214–219} Yet, large cardio-oncology trials are historically limited to assessment of cardiac effects alone. This may be, in part, due to the difficulty of assessing coronary vascular function in patients with cancer in vivo. The advancement of sophisticated imaging techniques (CMR, PET-CT, and so on) now allows for noninvasive assessment of coronary

vascular function,^{220,221} which could be used to assess chemotherapy-induced changes. Alternatively, considering that vascular dysfunction often manifests in the coronary and peripheral circulations simultaneously,^{222,223} measurement of peripheral vascular function (eg, flow-mediated dilation, pulse wave velocity, and laser Doppler flowmetry) may be a viable substitute with potential predictive capability for cardiac changes and adverse cardiovascular outcomes in cancer populations.^{67,224}

Conclusions

In conclusion, locoregional and nanoparticle-based delivery of anthracyclines remains commonplace in the treatment of various cancerous malignancies. However, whether they increase treatment efficacy and reduce cardiotoxicity compared with systemically administered anthracyclines is yet to be fully elucidated. Advancement in our understanding of anthracycline cardiotoxicity as a whole, as well as the role played by locoregional and nanoparticle-based administration techniques in reducing adverse cardiovascular events, will require continued investigation using a combination of preclinical and patient-based approaches. Ultimately, future work will largely benefit from close collaboration between physicians, researchers, and industry partners to ensure that patients treated with anthracycline-based regimens can receive ideal cancer care without significant risk to cardiovascular well-being.

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